

The electromagnetic spectrum

Student worksheet

Describe wave properties and calculate wavelengths relevant to radio astronomy. Show your reasoning and include units in each answer.

Name _____ Date _____

Wave properties

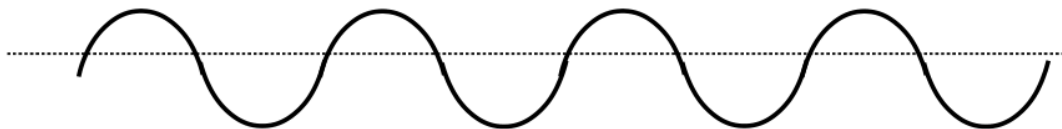
Mechanical waves need a medium, such as air or water. Electromagnetic waves can travel through a vacuum. Radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, and gamma rays are all electromagnetic waves.

Property	Meaning
Period T	Time for one oscillation, measured in seconds.
Frequency f	Oscillations per second, measured in hertz.
Amplitude A	Maximum size of the disturbance relative to equilibrium.
Wavelength λ	Distance between repeating points, measured in meters.
Wave speed v	Speed at which the disturbance travels.

Describe a wave

1. How are frequency and period related? Write an equation and explain its meaning.

2. Label the amplitude and wavelength on this snapshot of a wave. Mark the equilibrium level.

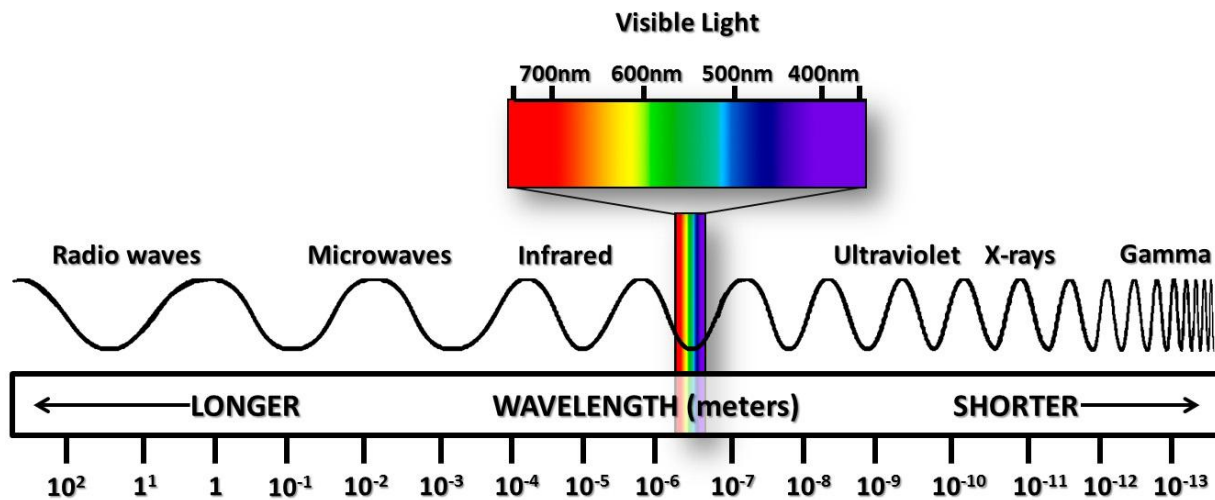


A wave shown as a function of position.

Connect speed and wavelength

Wave speed satisfies $v = f\lambda$. In a vacuum, electromagnetic waves travel at $c = 3.00 \times 10^8$ m/s. Therefore $c = f\lambda$. Higher frequency means shorter wavelength at the same speed.

Calculate and compare wavelengths



The electromagnetic spectrum spans a wide range of wavelengths.

Hydrogen emission

3. The horn telescope detects neutral hydrogen near 1420.4 MHz. Calculate the wavelength and express your answer in centimeters.

A radio frequency range

4. Consider signals from 87 MHz to 107 MHz. Calculate their wavelength range in meters. Which frequency has the longer wavelength?

An atomic scale

5. X-ray wavelengths can be comparable to atomic dimensions. Choose a reasonable atomic diameter or radius and give its value in meters. Calculate the frequency of light with that wavelength. Record your source for the atomic size.
